

LESSON 4-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Find the Whole**

Maryland Standard 6.RP.A.3c · Learning target – I can find the whole when I know a part and the percent it represents.

REMEMBER

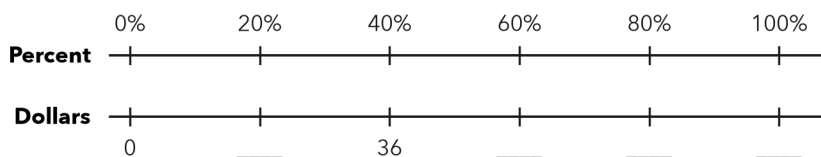
- The **part** is what you know. The **whole** is 100%.
- Step **down** to a friendly percent, then build **up** to 100%.
- Percent below 100%? The whole is **greater** than the part.

WORKED EXAMPLE

24 is 40% of what number?

Step 1 40% is 24, so 10% is $24 \div 4 = 6$.**Step 2** 100% is $10 \times 6 = 60$.**Step 3** Check: $0.4 \times 60 = 24$, so it checks.**Answer 60****WORD BANK****part** – the amount you know**whole** – the full amount; it is 100%**percent** – out of 100**double number line** – two lines that match amounts and percents**PRACTICE***show your work · label units*

- 1. Bike Helmet.** Andre has saved \$36. That is 40% of the cost of a bike helmet. Fill in the double number line. Then find the cost of the helmet.

Step 1 $40\% \div 2 = 20\%$. So $\$36 \div 2 =$ the 20% amount.**Step 2** Add that amount at each jump.**Step 3** The amount under 100% is the cost.**Cost** _____ dollars

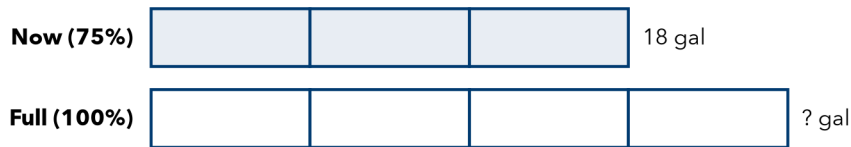
- 2.** Find each whole.

Step 1 Find 10% (or 5% or 25%) first.**Step 2** Multiply up to 100%.**a.** 13 is 10% of _____ **b.** 11 is 25% of _____ **c.** 21 is 30% of _____**d.** 32 is 40% of _____ **e.** 7 is 5% of _____ **f.** 54 is 60% of _____

3. **Water Tank.** A water tank is 75% full. It holds 18 gallons now. Each part of the tape diagram is 25% of the full tank. How many gallons does the tank hold when it is full?

Step 1 3 parts = 18 gallons. Find 1 part.

Step 2 The full tank is 4 parts.



Answer _____ gallons

4. **Chapter Book.** Lena has read 120 pages. That is **75%** of her book. How many pages are in the **whole** book?

Step 1 75% is 3 parts of 25%. Find 1 part: $120 \div 3$.

Step 2 Multiply by 4 to get 100%.

Answer _____ pages

5. **Reason.** 20 is 40% of number A. 20 is also 80% of number B. Without solving, which number is greater, A or B? Explain. Then find both numbers.

Frame ____ is greater because 20 is a ____ percent of it. $A = \underline{\hspace{1cm}}$ and $B = \underline{\hspace{1cm}}$.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about # ____

AT HOME Tell someone at home: if 10 is 50% of a number, what is the number? Explain how you know.

LESSON 4-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Solve for the Whole**

Maryland Standard 6.RP.A.3c • Learning target – I can write and solve an equation to find the whole from a part and a percent.

REMEMBER

- Write **part = percent × whole**, with w for the whole.
- Solve by dividing: $w = \text{part} \div \text{decimal}$.
- **Check:** decimal $\times$ whole should give the part.

WORKED EXAMPLE

\$18 is 60% of a shirt's first price. Find the first price.

Step 1 Write the equation: $0.6 \times w = 18$

Step 2 Divide: $w = 18 \div 0.6 = 180 \div 6 = 30$

Step 3 Check: $0.6 \times 30 = 18$, so it checks.

Answer \$30

WORD BANK

w – the letter for the whole you are looking for

equation – a math sentence with =

divide – undo multiplying

check – multiply back to get the part

PRACTICE

show your work • label units

1. Solve each equation for w .

Step 1 Divide the number on the right by the decimal.

Step 2 Check: decimal $\times w$ gives the number.

a. $0.2 \times w = 14$ _____ b. $0.25 \times w = 12$ _____ c. $0.8 \times w = 64$ _____

2. 33 is 55% of what number? Write an equation and solve it.

Step 1 Change 55% to a decimal.

Step 2 Use decimal $\times w = \text{part}$. The part is the number you know.

Step 3 Divide the part by the decimal.

a. Equation: _____

b. $w =$ _____

3. **Field Trip.** A class has \$315. That is **75%** of the trip cost. Find the total cost. How much **more** does the class need?

Step 1 Write $0.75 \times w = 315$ and divide.

Step 2 Subtract \$315 from the total.

Show your work.

Total and still needed _____

4. **Savings Goal.** Tasha has saved \$42. That is 35% of her goal. Write an equation and find her goal.

Equation and goal _____

5. **Find the Error.** Jamal solved “28 is 35% of w ” as $w = 28 \div 35 = 0.8$. Explain his mistake. Solve and check.

Frame Jamal’s mistake is ____, 35% = ____, so $w = 28 \div \text{__} = \text{__}$.

TEST PRACTICE

choose the best answer

6. 45 is 30% of a number w . Which equation can you use to find w ?

A $0.3 \times 45 = w$ **B** $0.3 \times w = 45$ **C** $45 \times w = 30$ **D** $w \div 45 = 0.3$

7. A team won 18 games. That is 60% of its games. How many games did it play?

Step 1 60% = 0.6. Write $0.6 \times w = 18$.

Step 2 Divide: $18 \div 0.6$.

A 10.8 **B** 24 **C** 30 **D** 78

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Explain to someone at home why the whole must be bigger than the part when the percent is less than 100%.